

The Geometry and Formatting of Reasoning Trajectories

To measure latent dynamics over the tail without the confounds of 2D projection, we define a scalar rotation statistic (`rotation_power`): the fraction of non-DC power in the period-6 bin of the 5280-dimensional state over a 24-unroll tail. This instrument is self-consistent to 4×10^{-8} , supports exact thresholds, and reproduces hand-assigned binary regime labels on 691 of 696 orbits (688/696 held out in leave-one-bank-out validation).

Using this instrument, we find that the recurrent map carries a sharp qualitative structure. A single instruction word switches the map between a settling regime and a damped period-6 rotation (e.g., “report the largest symbol” rotates 36/36 paired cells; “largest element” 0/36 at an identical token count). This switch is causally controlled by the token’s embedding row. Furthermore, the regime is memoryless in its parameter: swapping the injected state mid-trajectory re-aims the dynamics with no hysteresis in 62 of 64 switches, revealing a strong entering-versus-leaving asymmetry (entering costs median 15.5 unrolls; leaving costs 1.0).

However, behavioural attempts to connect this structure to capability revealed a massive evaluation confound. Scoring the first generated token, the model answers correctly on 12.5% of items. Reading the generated text, however, the answer is present in 78.1% (against a measured cross-item chance rate of 0.042 to 0.152). What occupies the first position is prose: the token “The” opens 39.4% of 1260 banked generations. Depth makes this worse rather than better—the rate of opening with “The” rises twentyfold from $r = 2$ to $r = 32$, while the rate of opening with the answer does not move ($p = 0.515$). The model is not failing to compute the answer; it is failing to put the answer where standard metrics look. Consequently, supplying the output format recovers the published benchmark number by two independent routes: 0.658 zero-shot under the original evaluation protocol, and 0.723 with five in-context examples.

The geometric and behavioural findings meet in a timing constraint. Measuring the rotation statistic over two distinct windows of the same orbit separates two facts: over the decision window (unrolls 0–11) rotating and settling prompts are indistinguishable (0.2292 against 0.2296, a separation of 0.0004), but over the tail they separate completely (0.862 against 0.298). Because the answer is best-ranked at a median unroll of four, behavioural nulls about the regime manipulate a property that has not yet come into existence when the readout is settled. This is a timing claim rather than a causal one: it does not show that the regime cannot affect behaviour, only that on this architecture the answer is fixed roughly twelve unrolls before the regime becomes measurable.

Finally, we note methodological limits. We withdrew several initial claims when pre-registered controls failed (e.g., a linear probe result was rejected when the instrument failed a deterministic positive control). We also note that the hypothesis that harder problems recruit more depth is architecturally untestable here. Huginn lacks a halting head; `num_steps = r` unrolls the whole sequence

together, meaning every position receives exactly r iterations with no mechanism to allocate compute by difficulty. Finally, our benchmark comparisons and the published figures share an uncorrected surface-form bias inherent to likelihood scoring.